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RAT IN THE SKULL

BY ROG PHILLIPS

Some people will be shocked by this story. Others will be deeply moved. Everyone who reads it will be talking about it. Read the first four pages: then put it down if you can.

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Dr. Joseph MacNare was not the sort of person one would expect him to be in the light of what happened. Indeed, it is safe to say that until the summer of 1955 he was more "normal", better adjusted, than the average college professor. And we have every reason to believe that he remained so, in spite of having stepped out of his chosen field.

At the age of thirty-four, he had to his credit a college textbook on advanced calculus, an introductory physics, and seventy-two papers that had appeared in various journals, copies of which were in neat order in a special section of the bookcase in his office at the university, and duplicate copies of which were in equally neat order in his office at home. None of these were in the field of psychology, the field in which he was shortly to become famous—or infamous. But anyone who studies the published writings of Dr. MacNare must inevitably conclude that he was a competent, responsible scientist, and a firm believer in institutional research, research by teams, rather than in private research and go-it-alone secrecy, the course he eventually followed.

In fact, there is every reason to believe he followed this course with the greatest of reluctance, aware of its pitfalls, and that he took every precaution that was humanly possible.

Certainly, on that day in late August, 1955, at the little cabin on the Russian River, a hundred miles upstate from the university, when Dr. MacNare completed his paper on *An Experimental Approach to the Psychological Phenomena of Verification*, he had no slightest thought of "going it alone."

It was mid-afternoon. His wife, Alice, was dozing on the small dock that stretched out into the water, her slim figure tanned a smooth brown that was just a shade lighter than her hair. Their eight-year-old son, Paul, was fifty yards upstream playing with some other boys, their shouts the only sound except for the whisper of rushing water and the sound of wind in the trees.

Dr. MacNare, in swim trunks, his lean muscular body hardly tanned at all, emerged from the cabin and came out on the dock.

"Wake up, Alice," he said, nudging her with his foot. "You have a husband again."

"Well, it's about time," Alice said, turning over on her back and looking up at him, smiling in answer to his happy grin.

He stepped over her and went out on the diving board, leaping up and down on it, higher and higher each time, in smooth coördination, then went into a one and a half gainer, his body cutting into the water with a minimum of splash.

His head broke the surface. He looked up at his wife, and laughed in the sheer pleasure of being alive. A few swift strokes brought him to the foot of the ladder. He climbed, dripping water, to the dock, then sat down by his wife.

"Yep, it's done," he said. "How many days of our vacation left? Two? That's time enough for me to get a little tan. Might as well make the most of it. I'm going to be working harder this winter than I ever did in my life."

"But I thought you said your paper was done!"

"It is. But that's only the beginning. Instead of sending it in for publication, I'm going to submit it to the directors, with a request for facilities and personnel to conduct a line of research based on pages twenty-seven to thirty-two of the paper."

"And you think they'll grant your request?"

"There's no question about it," Dr. MacNare said, smiling confidently. "It's the most important line of research ever opened up to experimental psychology. They'll be forced to grant my request. It will put the university on the map!"

Alice laughed, and sat up and kissed him.

"Maybe they won't agree with you," she said. "Is it all right for me to read the paper?"

"I wish you would," he said. "Where's that son of mine? Upstream?" He leaped to his feet and went to the diving board again.

"Better walk along the bank, Joe. The stream is too swift."

"Nonsense!" Dr. MacNare said.

He made a long shallow dive, then began swimming in a powerful crawl that took him upstream slowly. Alice stood on the dock watching him until he was lost to sight around the bend, then went into the cabin. The completed paper lay beside the typewriter.

Alice had her doubts. "I'm not so sure the board will approve of this," she said. Dr. MacNare, somewhat exasperated, said, "What makes you think that? Pavlov experimented with his dog, physiological experiments with rats, rabbits, and other animals go on all the time. There's nothing cruel about it."

"Just the same...." Alice said. So Dr. MacNare cautiously resisted the impulse to talk about his paper with his fellow professors and his most intelligent students. Instead, he merely turned his paper in to the board at the earliest opportunity and kept silent, waiting for their decision.

He hadn't long to wait. On the last Friday of September he received a note requesting his presence in the board room at three o'clock on Monday. He rushed home after his last class and told Alice about it.

"Let's hope their decision is favorable," she said.

"It has to be," Dr. MacNare answered with conviction.

He spent the week-end making plans. "They'll probably assign me a machinist and a couple of electronics experts from the hill," he told Alice. "I can use graduate students for work with the animals. I hope they give me Dr. Munitz from Psych as a consultant, because I like him much better than Veerhof. By early spring we should have things rolling."

Monday at three o'clock on the dot, Dr. MacNare knocked on the door of the board room, and entered. He was not unfamiliar with it, nor with the faces around the massive walnut conference table. Always before he had known what to expect—a brief commendation for the revisions in his textbook on calculus for its fifth printing, a nice speech from the president about his good work as a prelude to a salary raise—quiet, expected things. Nothing unanticipated had ever happened here.

Now, as he entered, he sensed a difference. All eyes were fixed on him, but not with admiration or friendliness. They were fixed more in the manner of a restaurateur watching the approach of a cockroach along the surface of the counter.

Suddenly the room seemed hot and stuffy. The confidence in Dr. MacNare's expression evaporated. He glanced back toward the door as though wishing to escape.

"So it's *you*!" the president said, setting the tone of what followed.

"This is *yours*?" the president added, picking up the neatly typed manuscript, glancing at it, and dropping it back on the table as though it were something unclean.

Dr. MacNare nodded, and cleared his throat nervously to say yes, but didn't get the chance.

"We—all of us—are amazed and shocked," the president said. "Of course, we understand that psychology is not your field, and you probably were thinking only from the mathematical viewpoint. We are agreed on that. What you propose, though...." He shook his head slowly. "It's not only out of the question, but I'm afraid I'm going to have to request that you forget the whole thing—put this paper where no one can see it, preferably destroy it. I'm sorry, Dr. MacNare, but the university simply cannot afford to be associated with such a thing even remotely. I'll put it bluntly because I feel strongly about it, as do the other members of the Board. *If this paper is published or in any way comes to light, we will be forced to request your resignation from the faculty.*"

"But why?" Dr. MacNare asked in complete bewilderment.

"Why?" another board member exploded, slapping the table. "It's the most inhuman thing I ever heard of, strapping a newborn animal onto some kind of frame and tying its legs to control levers, with the intention of never letting it free. The most fiendish and inhuman torture imaginable! If you didn't have such an outstanding record I would be for demanding your resignation at once."

"But that's not true!" Dr. MacNare said. "It's not torture! Not in any way! Didn't you read the paper? Didn't you understand that—"

"I read it," the man said. "We all read it. Every word."

"Then you should have understood—" Dr. MacNare said.

"We read it," the man repeated, "and we discussed some aspects of it with Dr. Veerhof without bringing your paper into it, nor your name."

"Oh," Dr. MacNare said. "Veerhof...."

"He says experiments, very careful experiments, have already been conducted along the lines of getting an animal to understand a symbol system and it can't be done. The nerve paths aren't there. Your line of research, besides being inhumanly cruel, would accomplish nothing."

"Oh," Dr. MacNare said, his eyes flashing. "So you know all about the results of an experiment in an untried field without performing the experiments!"

"According to Dr. Veerhof that field is not untried but rather well explored," the board member said. "Giving an animal the means to make vocal sounds would not enable it to form a symbol system."

"I disagree," Dr. MacNare said, seething. "My studies indicate clearly—"

"I think," the president said with a firmness that demanded the floor, "our position has been made very clear, Dr. MacNare. The matter is now closed. Permanently. I hope you will have the good sense, if I may use such a strong term, to forget the whole thing. For the good of your career and your very nice wife and son. That is all." He held the manuscript toward Dr. MacNare.

"I can't understand their attitude!" Dr. MacNare said to Alice when he told her about it.

"Possibly I can understand it a little better than you, Joe," Alice said thoughtfully. "I had a little of what I think they feel, when I first read your paper. A—a prejudice against the idea of it, is as closely as I can describe it. Like it would be violating the order of nature, giving an animal a soul, in a way."

"Then you feel as they do?" Dr. MacNare said.

"I didn't say that, Joe." Alice put her arms around her husband and kissed him fiercely. "Maybe I feel just the opposite, that if there is some way to give an animal a soul, we should do it."

Dr. MacNare chuckled. "It wouldn't be quite that cosmic. An animal can't be given something it doesn't have already. All that can be done is to give it the means to fully capitalize on what it has. Animals—man included—can only do by observing the results. When you move a finger, what you really do is send a neural impulse out from the brain along one particular nerve or one particular set of nerves, but you can never learn that, nor just what it is you do. All that you can know is that when you do a definite *something* your eyes and sense of touch bring you the information that your finger moved. But if that finger were attached to a voice element that made the sound *ah*, and you could never see your finger, all you could ever know is that when you did that particular *something* you made a certain vocal sound. Changing the resultant effect of mental commands to include things normally impossible to you may expand the potential of your mind, but it won't give you a soul if you don't have one to begin with."

"You're using Veerhof's arguments on me," Alice said. "And I think we're arguing from separate definitions of a soul. I'm afraid of it, Joe. It would be a tragedy, I think, to give some animal—a rat, maybe—the soul of a poet, and then have it discover that it is only a rat."

"Oh," Dr. MacNare said. "*That* kind of soul. No, I'm not that optimistic about the results. I think we'd be lucky to get any results at all, a limited vocabulary that the animal would use meaningfully. But I do think we'd get that."

"It would take a lot of time and patience."

"And we'd have to keep the whole thing secret from everyone," Dr. MacNare said. "We couldn't even let Paul have an inkling of it, because he might say something to one of his playmates, and it would get back to some member of the board. How could we keep it secret from Paul?"

"Paul knows he's not allowed in your study," Alice said. "We could keep everything there—and keep the door locked."

"Then it's settled?"

"Wasn't it, from the very beginning?" Alice put her arms around her husband and her cheek against his ear to hide her worried expression. "I love you, Joe. I'll help you in any way I can. And if we haven't enough in the savings account, there's always what Mother left me."

"I hope we won't have to use any of it, sweetheart," he said.

The following day Dr. MacNare was an hour and a half late coming home from the campus. He had been, he announced casually, to a pet store.

"We'll have to hurry," said Alice. "Paul will be home any minute."

She helped him carry the packages from the car to the study. Together they moved things around to make room for the gleaming new cages with their white rats and hamsters and guinea pigs. When it was done they stood arm in arm viewing their new possession.

To Alice MacNare, just the presence of the animals in her husband's study brought the research project into reality. As the days passed that romantic feeling became fact.

"We're going to have to do together," Joe MacNare told her at the end of the first week, "what a team of a dozen specialists in separate fields should be doing. Our first job, before we can do anything else, is to study the natural movements of each species and translate them into patterns of robot directives."

"Robot directives?"

"I visualize it this way," Dr. MacNare said. "The animal will be strapped comfortably in a frame so that its body can't move but its legs can. Its legs will be attached to four separate, free-moving levers which make a different electrical contact for every position. Each electrical contact, or control switch, will cause the robot body to do one specific thing, such as move a leg, utter some particular sound through its voice box, or move just one finger. Can you visualize that, Alice?"

Alice nodded.

"Okay. Now, one leg has to be used for nothing but voice sounds. That leaves three legs for control of the movements of the robot body. In body movement there will be simultaneous movements and sequences. A simple sequence can be controlled by one leg. All movements of the robot will have to be reduced to not more than three concurrent sequences of movement of the animal's legs. Our problem, then, is to make the unlearned and the most natural movements of the legs of the animal control the robot body's movements in a functional manner."

Endless hours were consumed in this initial study and mapping. Alice worked at it while her husband was at the university and Paul was at school. Dr. MacNare rushed home each day to go over what she had done and continue the work himself.

He grew more and more grudging of the time his classes took. In December he finally wrote to the three technical journals that had been expecting papers from him for publication during the year that he would be too busy to do them.

By January the initial phase of research was well enough along so that Dr. MacNare could begin planning the robot. For this he set up a workshop in the garage.

In early February he finished what he called the "test frame." After Paul had gone to bed, Dr. MacNare brought the test frame into the study from the garage. To Alice it looked very much like the insides of a radio.

She watched while he placed a husky-looking male white rat in the body harness fastened to the framework of aluminum and tied its legs to small metal rods.

Nothing happened except that the rat kept trying to get free, and the small metal rods tied to its feet kept moving in pivot sockets.

"Now!" Dr. MacNare said excitedly, flicking a small toggle switch on the side of the assembly.

Immediately a succession of vocal sounds erupted from the speaker. They followed one another, making no sensible word.

"*He's* doing that," Dr. MacNare said triumphantly.

"If we left him in that, do you think he'd eventually associate his movements with the sounds?"

"It's possible. But that would be more on the order of what we do when we drive a car. To some extent a car becomes an extension of the body, but you're always aware that your hands are on the steering wheel, your foot on the gas pedal or brake. You extend your awareness consciously. You interpret a slight tremble in the steering wheel as a shimmy in the front wheels. You're oriented primarily to your body and only secondarily to the car as an extension of you."

Alice closed her eyes for a moment. "Mm hm," she said.

"And that's the best we could get, using a rat that knows already it's a rat."

Alice stared at the struggling rat, her eyes round with comprehension, while the loudspeaker in the test frame said, "Ag-pr-ds-raf-os-dg...."

Dr. MacNare shut off the sound and began freeing the rat.

"By starting with a newborn animal and never letting it know what it is," he said, "we can get a complete extension of the animal into the machine, in its orientation. So complete that if you took it out of the machine after it grew up, it would have no more idea of what had happened than—than your brain if it were taken out of your head and put on a table!"

"Now I'm getting that *feeling* again, Joe," Alice said, laughing nervously. "When you said that about my brain I thought, 'Or my soul?'"

Dr. MacNare put the rat back in its cage.

"There might be a valid analogy there," he said slowly. "If we have a soul that survives after death, what is it like? It probably interprets its surroundings in terms of its former orientation in the body."

"That's a little of what I mean," Alice said. "I can't help it, Joe. Sometimes I feel so sorry for whatever baby animal you'll eventually use, that I want to cry. I feel so sorry for it, because *we will never dare let it know what it really is*!"

"That's true. Which brings up another line of research that should be the work of one expert on the team I ought to have for this. As it is, I'll turn it over to you to do while I build the robot."

"What's that?"

"Opiates," Dr. MacNare said. "What we want is an opiate that can be used on a small animal every few days, so that we can take it out of the robot, bathe it, and put it back again without its knowing about it. There probably is no ideal drug. We'll have to test the more promising ones."

Later that night, as they lay beside each other in the silence and darkness of their bedroom, Dr. MacNare sighed deeply.

"So many problems," he said. "I sometimes wonder if we can solve them all. *See* them all...."

To Alice MacNare, later, that night in early February marked the end of the first phase of research—the point where two alternative futures hung in the balance, and either could have been taken. That night she might have said, there in the darkness, "Let's drop it," and her husband might have agreed.

She thought of saying it. She even opened her mouth to say it. But her husband's soft snores suddenly broke the silence of the night. The moment of return had passed.

Month followed month. To Alice it was a period of rushing from kitchen to hypodermic injections to vacuum cleaner to hypodermic injections, her key to the study in constant use.

Paul, nine years old now, took to spring baseball and developed an indifference to TV, much to the relief of both his parents.

In the garage workshop Dr. MacNare made parts for the robot, and kept a couple of innocent projects going which he worked on when his son Paul evinced his periodic curiosity about what was going on.

Spring became summer. For six weeks Paul went to Scout camp, and during those six weeks Dr. MacNare reorganized the entire research project in line with what it would be in the fall. A decision was made to use only white rats from then on. The rest of the animals were sold to a pet store, and a system for automatically feeding, watering, and keeping the cages clean was installed in preparation for a much needed two weeks' vacation at the cabin.

When the time came to go, they had to tear themselves away from their work by an effort of will—aided by the realization that they could get little done with Paul underfoot.

September came all too soon. By mid-September both Dr. MacNare and his wife felt they were on the home stretch. Parts of the robot were going together and being tested, the female white rats were being bred at the rate of one a week so that when the robot was completed there would be a supply of newborn rats on hand.

October came, and passed. The robot was finished, but there were minor defects in it that had to be corrected.

"Adam," Dr. MacNare said one day, "will have to wear this robot all his life. It has to be just right."

And with each litter of baby rats Alice said, "I wonder which one is Adam."

They talked of Adam often now, speculating on what he would be like. It was almost, they decided, as though Adam were their second child.

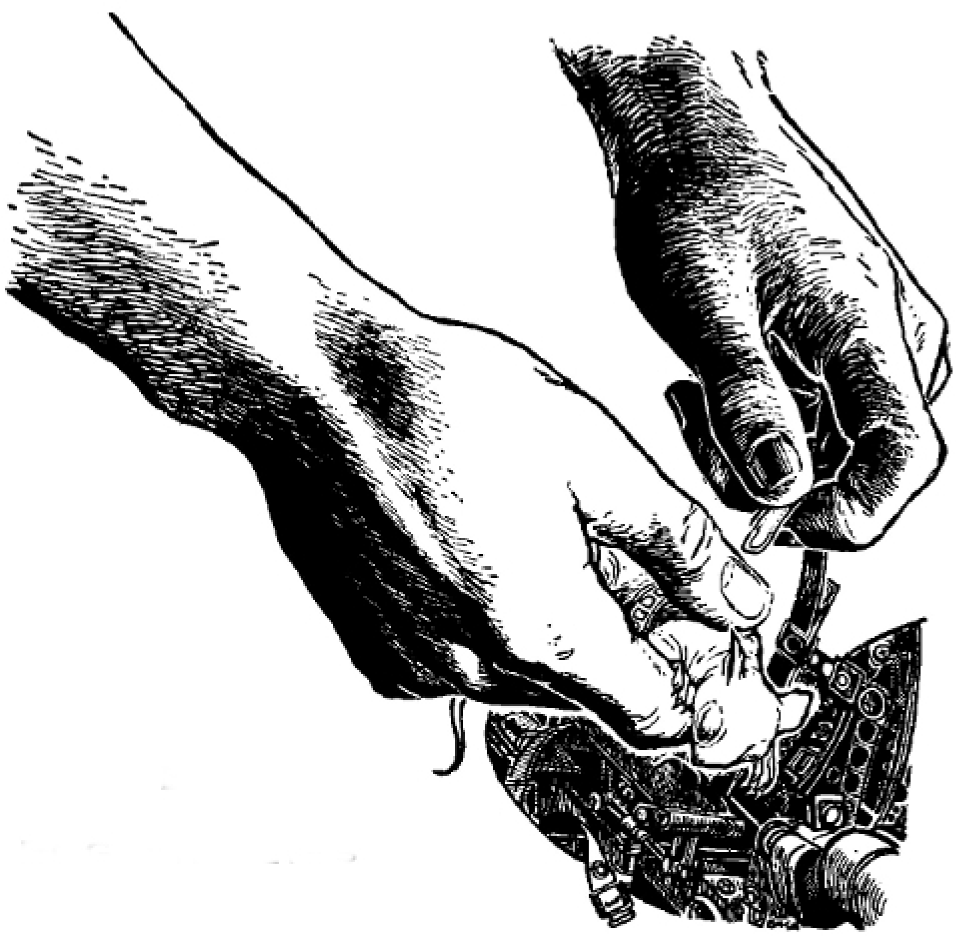
And finally, on November 2, 1956, everything was ready. Adam would be born in the next litter, due in about three days.

The amount of work that had gone into preparation for the great moment is beyond conception. Four file cabinet drawers were filled with notes. By actual measurement seventeen feet of shelf space was filled with books on the thousand and one subjects that had to be mastered. The robot itself was a masterpiece of engineering that would have done credit to the research staff of a watch manufacturer. The vernier adjustments alone, used to compensate daily for the rat's growth, had eight patentable features.

And the skills that had had to be acquired! Alice, who had never before had a hypodermic syringe in her hand, could now inject a precisely measured amount of opiate into the tiny body of a baby rat with calm confidence in her skill.

After such monumental preparation, the great moment itself was anticlimactic. While the mother of Adam was still preoccupied with the birth of the remainder of the brood, Adam, a pink helpless thing about the size of a little finger, was picked up and transfered to the head of the robot.

His tiny feet, which he would never know existed, were fastened with gentle care to the four control rods. His tiny head was thrust into a helmet attached to a pivot-mounted optical system, ending in the lenses that served the robot for eyes. And finally a transparent plastic cover contoured to the shape of the back of a human head was fastened in place. Through it his feeble attempts at movement could be easily observed.



Thus, Dr. MacNare's Adam was born into his body, and the time of the completion of his birth was one-thirty in the afternoon on the fifth day of November, 1956.

In the ensuing half hour all the cages of rats were removed from the study, the floor was scrubbed, and deodorizers were sprayed, so that no slightest trace of Adam's lowly origins remained. When this was done, Dr. MacNare loaded the cages into his car and drove them to a pet store that had agreed to take them.

When he returned, he joined Alice in the study, and at five minutes before four, with Alice hovering anxiously beside him, he opened the cover on Adam's chest and turned on the master switch that gave Adam complete dominion over his robot body.

Adam was beautiful—and monstrous. Made of metal from the neck down, but shaped to be covered by padding and skin in human semblance. From the neck up the job was done. The face was human, masculine, handsome, much like that of a clothing store dummy except for its mobility of expression, and the incongruity of the rest of the body.

The voice-control lever and contacts had been designed so that the ability to produce most sounds would have to be discovered by Adam as he gained control of his natural right front leg. Now the only sounds being uttered were *oh*, *ah*, *mm*, and *ll*, in random order. Similarly, the only movements of his arms and legs were feeble, like those of a human baby. The tremendous strength in his limbs was something he would be unable to tap fully until he had learned conscious coördination.

After a while Adam became silent and without movement. Alarmed, Dr. MacNare opened the instrument panel in the abdomen. The instruments showed that Adam's pulse and respiration were normal. He had fallen asleep.

Dr. MacNare and his wife stole softly from the study, and locked the door.

After a few days, with the care and feeding of Adam all that remained of the giant research project, the pace of the days shifted to that of long-range patience.

"It's just like having a baby," Alice said.

"You know something?" Dr. MacNare asked. "I've had to resist passing out cigars. I hate to say it, but I'm prouder of Adam than I was of Paul when he was born."

"So am I, Joe," Alice said quietly. "But I'm getting a little of that scared feeling back again."

"In what way?"

"He watches me. Oh, I know it's natural for him to, but I do wish you had made the eyes so that his own didn't show as little dark dots in the center of the iris."

"It couldn't be helped," Dr. MacNare said. "He has to be able to see, and I had to set up the system of mirrors so that the two axes of vision would be three inches apart as they are in the average human pair of eyes."

"Oh, I know," said Alice. "Probably it's just something I've seized on. But when he watches me, I find myself holding my breath in fear that he can read in my expression the secret we have to keep from him, that he is a rat."

"Forget it, Alice. That's outside his experience and beyond his comprehension."

"I know," Alice sighed. "When he begins to show some of the signs of intelligence a baby has, I'll be able to think of him as a human being."

"Sure, darling," Dr. MacNare said.

"Do you think he ever will?"

"That," Dr. MacNare said, "is the big question. I think he will. I think so now even more than I did at the start. Aside from eating and sleeping, he has no avenue of expression except his robot body, and *no source of reward except that of making sense—human sense*."

The days passed, and became weeks, then months. During the daytime when her husband was at the university and her son was at school, Alice would spend most of her hours with Adam, forcing herself to smile at him and talk to him as she had to Paul when he was a baby. But when she watched his motions through the transparent back of his head, his leg motions remained those of attempted walking and attempted running.

Then, one day when Adam was four months old, things changed—as abruptly as the turning on of a light.

The unrewarding walking and running movements of Adam's little legs ceased. It was evening, and both Dr. MacNare and his wife were there.

For a few seconds there was no sound or movement from the robot body. Then, quite deliberately, Adam said, "Ah."

"Ah," Dr. MacNare echoed. "Mm, Mm, ah. Ma-ma."

"Mm," Adam said.

The silence in the study became absolute. The seconds stretched into eternities. Then—

"Mm, ah," Adam said. "Mm, ah."

Alice began crying with happiness.

"Mm, ah," Adam said. "Mm, ah. Ma-ma. Mamamamama."

Then, as though the effort had been too much for Adam, he went to sleep.

Having achieved the impossible, Adam seemed to lose interest in it. For two days he uttered nothing more than an occasional involuntary syllable.

"I would call that as much of an achievement as speech itself," Dr. MacNare said to his wife. "His right front leg has asserted its independence. If each of his other three legs can do as well, he can control the robot body."

It became obvious that Adam was trying. Though the movements of his body remained non-purposive, the pauses in those movements became more and more pregnant with what was obviously mental effort.

During that period there was of course room for argument and speculation about it, and even a certain amount of humor. Had Adam's right front leg, at the moment of achieving meaningful speech, suffered a nervous breakdown? What would a psychiatrist have to say about a white rat that had a nervous breakdown in its right front leg?

"The worst part about it," Dr. MacNare said to his wife, "is that if he fails to make it he'll have to be killed. He can't have permanent frustration forced onto him, and, by now, returning him to his natural state would be even worse."

"And he has such a stout little heart," Alice said. "Sometimes when he looks at me I'm sure he knows what is happening and he wants me to know he's trying."

When they went to bed that night they were more discouraged than they had ever been.

Eventually they slept. When the alarm went off, Alice slipped into her robe and went into the study first, as she always did.

A moment later she was back in the bedroom, shaking her husband's shoulder.

"Joe!" she whispered. "Wake up! Come into the study!"

He leaped out of bed and rushed past her. She caught up with him and pulled him to a stop.

"Take it easy, Joe," she said. "Don't alarm him."

"Oh." Dr. MacNare relaxed. "I thought something had happened."

"Something has!"

They stopped in the doorway of the study. Dr. MacNare sucked in his breath sharply, but remained silent.

Adam seemed oblivious of their presence. He was too interested in something else.

He was interested in his hands. He was holding his hands up where he could see them, and he was moving them independently, clenching and unclenching the metal fingers with slow deliberation.

Suddenly the movement stopped. He had become aware of them. Then, impossibly, unbelievably, he spoke.

"Ma ma," Adam said. Then, "Pa pa."

"Adam!" Alice sobbed, rushing across the study to him and sinking down beside him. Her arms went around his metal body. "Oh, Adam," she cried happily.

It was the beginning. The date of that beginning is not known. Alice MacNare believes it was early in May, but more probably it was in April. There was no time to keep notes. In fact, there was no longer a research project nor any thought of one. Instead, there was Adam, the person. At least, to Alice he became that, completely. Perhaps, also, to Dr. MacNare.

Dr. MacNare quite often stood behind Adam where he could watch the rat body through the transparent skull case while Alice engaged Adam's attention. Alice did the same, at times, but she finally refused to do so any more. The sight of Adam the rat, his body held in a net attached to the frame, his head covered by the helmet, his four legs moving independently of one another with little semblance of walking or running motion nor even of coördination, but with swift darting motions and pauses pregnant with meaning, brought back to Alice the old feeling of vague fear, and a tremendous surge of pity for Adam that made her want to cry.

Slowly, subtly, Adam's rat body became to Alice a pure brain, and his legs four nerve ganglia. A brain covered with short white fur; and when she took him out of his harness under opiate to bathe him, she bathed him as gently and carefully as any brain surgeon sponging a cortical surface.

Once started, Adam's mental development progressed rapidly. Dr. MacNare began making notes again on June 2, 1957, just ten days before the end, and it is to these notes that we go for an insight into Adam's mind.

On June 4th Dr. MacNare wrote, "I am of the opinion that Adam will never develop beyond the level of a moron, in the scale of human standards. He would probably make a good factory worker or chauffeur, in a year or two. But he is consciously aware of himself as Adam, he thinks in words and simple sentences with an accurate understanding of their meaning, and he is able to do new things from spoken instructions. There is no question, therefore, but that he has an integrated mind, entirely human in every respect."

On June 7th Dr. MacNare wrote, "Something is developing which I hesitate to put down on paper—for a variety of reasons. Creating Adam was a scientific experiment, nothing more than that. Both the premises on which the project was based have been proven: that the principle of verification is the main factor in learned response, and that, given the proper conditions, some animals are capable of abstract symbol systems and therefore of thinking with words to form meaningful concepts.

"Nothing more was contemplated in the experiment. I stress this because—Adam is becoming deeply religious—and before any mistaken conclusions are drawn from this I will explain what caused this development. It was an oversight of a type that is bound to happen in any complex project.

"Alice's experimental data on the effects of opiates, and especially the data on increasing the dose to offset growing tolerance, were based on observation of the subject alone, without any knowledge of the mental aspects of increased tolerance—which would of course be impossible except with human subjects.

"Unknown to us, Adam has been becoming partly conscious during his bath. Just conscious enough to be vaguely aware of certain sensations, and to remember them afterward. Few, if any, of these half remembered sensations are such that he can fit them into the pattern of his waking reality.

"The one that has had the most pronounced influence on him is, to quote him, 'Feel clean inside. Feel good.' Quite obviously this sensation is caused by his bath.

"With it is a distinct feeling of disembodiment, of being—and these are his own words—'outside my body'! This, of course, is an accurate realization, because to him the robot is his body, and he knows nothing of the existence of his actual, living, rat body.

"In addition to these two effects, there is a third one. A feeling of walking, and sometimes of floating, of stumbling over things he can't see, of talking, of being talked to by disembodied voices.

"The explanation of this is also obvious. When he is being bathed his legs are moved about. Any movement of a leg is to him either a spoken sound or a movement of some part of his robot body. Any movement of his right front leg, for example, tells his mind that he is making a sound. But, since his leg is not connected to the sound system of his robot body, his ears bring no physical verification of the sound. The mental anticipation of that verification then becomes a disembodied voice to him.

"The end result of all this is that Adam is becoming convinced that there is a hidden side of things (which there is), and that it is supernatural (which it is, *in the framework of his orientation*).

"What we are going to have to do is make sure he is completely unconscious before taking him out and bathing him. His mental health is far more important than exploring the interesting avenues opened up by this unforeseen development.

"I do intend, however, to make one simple test, while he is fully awake, before dropping this avenue of investigation."

Dr. MacNare does not state in his notes what this test was to be: but his wife says that it probably refers to the time when he pinched Adam's tail and Adam complained of a sudden, violent headache. This transference is the one well known to doctors. Unoriented pain in the human body manifests itself as a "headache," when the source of the pain is actually the stomach, or the liver, or any one of a hundred spots in the body.

The last notes made by Dr. MacNare were those of June 11, 1957, and are unimportant except for the date. We return, therefore, to actual events, so far as they can be reconstructed.

We have said little or nothing about Dr. MacNare's life at the university after embarking on the research project, nor of the social life of the MacNares. As conspirators, they had kept up their social life to avoid any possibility of the board getting curious about any radical change in Dr. MacNare's habits; but as time went on both Dr. MacNare and his wife became so engrossed in their project that only with the greatest reluctance did they go anywhere.

The annual faculty party at Professor Long's on June 12th was something they could not evade. Not to have gone would have been almost tantamount to a resignation from the university.

"Besides," Alice had said when they discussed the matter in May, "isn't it about time to do a little hinting that you have something up your sleeve?"

"I don't know, Alice," Dr. MacNare had said. Then a smile quirked his lips and he said, "I wouldn't mind telling off Veerhof. I've never gotten over his deciding something was impossible without enough data to pass judgment." He frowned. "We are going to have to let the world know about Adam pretty soon, aren't we? That's something I haven't thought about. But not yet. Next fall will be time enough."

"Don't forget, Joe," Alice said at dinner. "Tonight's the party at Professor Long's."

"How can I forget with you reminding me?" Dr. MacNare said, winking at his son.

"And you, Paul," Alice said. "I don't want you leaving the house. You understand? You can watch TV, and I want you in bed by nine thirty."

"Ah, Mom!" Paul protested. "Nine thirty?" He suppressed a grin. He had a party of his own planned.

"And you can wipe the dishes for me. We have to be at Professor Long's by eight o'clock."

"I'll help you," Dr. MacNare said.

"No, you have to get ready. Besides don't you have to look up something for one of the faculty?"

"I'd forgotten," said Dr. MacNare. "Thanks for reminding me."

After dinner he went directly to the study. Adam was sitting on the floor playing with his wooden blocks. They were alphabet blocks, but he didn't know that yet. The summer project was going to be teaching him the alphabet. Already, though, he preferred placing them in straight rows rather than stacking them up.

At seven o'clock Alice rapped on the door to the study.

"Time to get dressed, Joe," she called.

"You'll be all right while we're gone, Adam?" Dr. MacNare said.

"I be all right, papa," Adam said. "I sleep."

"That's good," Dr. MacNare said. "I'll turn out the light."

At the door he waited until Adam had sat down in the chair he always slept on, and settled himself. Then he pushed the switch just to the right of the door and went out.

"Hurry, dear," Alice called.

"I'm hurrying," Dr. MacNare protested—and, for the first time, he forgot to lock the study door.

The bathroom was next to the study, the wall between them soundproofed by a ceiling-high bookshelf in the study filled with thousands of books. On the other side was the master bedroom, with a closet with sliding panels that opened both on the bedroom and the bathroom. These sliding panels were partly open, so that Dr. MacNare and Alice could talk.

"Did you lock the study door?"

"Of course," Dr. MacNare said. "But I'll check before we leave."

"How is Adam taking being alone tonight?" Alice called.

"Okay," Dr. MacNare said. "Damn!"

"What's the matter, Joe?"

"I forgot to get razor blades."

The conversation died down.

Alice MacNare finished dressing.

"Aren't you ready yet, Joe?" she called. "It's almost a quarter to eight."

"Be right with you. I nicked myself shaving with an old blade. The bleeding's almost stopped now."

Alice went into the living room. Paul had turned on the TV and was sprawled out on the rug.

"You be sure and stay home, and be in bed by nine thirty, Paul," she said. "Promise?"

"Ah, Mom," he protested. "Well, all right."

Dr. MacNare came into the room, still working on his tie. A moment later they went out the front door. They had been gone less than five minutes when there was a knock. Paul jumped to his feet and opened the door.

"Hi, Fred, Tony, Bill," he said.

The boys, all nine years old, sprawled on the rug and watched television. It became eight o'clock, eight thirty, and finally five minutes to nine. The commercial began.

"Where's your bathroom?" Tony asked.

"In there," Paul said, pointing vaguely at the doorway to the hall.

Tony got up off the floor and went into the hall. He saw several doors, all looking much alike. He picked one and opened it. It was dark inside. He felt along the wall for a light switch and found it. Light flooded the room. He stared at what he saw for perhaps ten seconds, then turned and ran down the hall to the living room.

"Say, Paul!" he said. "You never said anything about having a real honest to gosh robot!"

"What are you talking about?" Paul said.

"In that room in there!" Tony said. "Come on. I'll show you!"

The TV program forgotten, Paul, Fred, and Bill crowded after him. A moment later they stood in the doorway to the study, staring in awe at the strange figure of metal that sat motionless in a chair across the room.

Adam, it seems certain, was asleep, and had not been wakened by this intrusion nor the turning on of the light.

"Gee!" Paul said. "It belongs to Dad. We'd better get out of here."

"Naw," Tony said with a feeling of proprietorship at having been the original discoverer. "Let's take a look. He'll never know about it."

They crossed the room slowly, until they were close up to the robot figure, marveling at it, moving around it.

"Say!" Bill whispered, pointing. "What's that in there? It looks like a white rat with its head stuck into that kind of helmet thing."

They stared at it a moment.

"Maybe it's dead. Let's see."

"How you going to find out?"

"See those hinges on the cover?" Tony said importantly. "Watch." With cautious skill he opened the transparent back half of the dome, and reached in, wrapping his fingers around the white rat.

He was unable to get it loose, but he succeeded in pulling its head free of the helmet.

At the same time Adam awoke.

"Ouch!" Tony cried, jerking his hand away. "He bit me!"

"He's alive all right," Bill said. "Look at him glare!" He prodded the body of the rat and pulled his hand away quickly as the rat lunged.

"Gee, look at its eyes," Paul said nervously. "They're getting blood-shot."

"Dirty old rat!" Tony said vindictively, jabbing at the rat with his finger and evading the snapping teeth.

"Get its head back in there!" Paul said desperately. "I don't want papa to find out we were in here!" He reached in, driven by desperation, pressing the rat's head between his fingers and forcing it back into the tight fitting helmet.

Immediately screaming sounds erupted from the lips of the robot. (It was determined by later examination that only when the rat's body was completely where it should be were the circuits operable.)

"Let's get out of here!" Tony shouted, and dived for the door, thereby saving his life.

"Yeah! Let's get out of here!" Fred shouted as the robot figure rose to its feet. Terror enabled him to escape.

Bill and Paul delayed an instant too long. Metal fingers seized them. Bill's arm snapped halfway between shoulder and elbow. He screamed with pain and struggled to free himself.

Paul was unable to scream. Metal fingers gripped his shoulder, with a metal thumb thrust deeply against his larynx, paralyzing his vocal cords.

Fred and Tony had run into the front room. There they waited, ready to start running again. They could hear Bill's screams. They could hear a male voice jabbering nonsense, and finally repeating over and over again, "Oh my, oh my, oh my," in a tone all the more horrible because it portrayed no emotion whatever.

Then there was silence.

The silence lasted several minutes. Then Bill began to sniffle, rubbing his knuckles in his eyes. "I wanta go home," he whimpered.

"Me too."

They took each other's hand and tiptoed to the front door, watching the open doorway to the hall. When they reached the front door Tony opened it, and when it was open they ran, not stopping to close the door behind them.

There isn't much more to tell. It is known that Tony and Bill arrived at their respective homes, saying nothing of what had happened. Only later did they come forward and admit their share in the night's events.

Joe and Alice MacNare arrived home from the party at Professor Long's at twelve thirty, finding the front door wide open, the lights on in the living room, and the television on.

Sensing that something was wrong, Alice hurried to her son's room and discovered he wasn't there. While she was doing that, Joe shut the front door and turned off the television.

Alice returned to the living room, eyes round with alarm, and said, "Paul's not in his room!"

"Adam!" Joe croaked, and rushed into the hallway, with Alice following more slowly.

She reached the open door of the study in time to see the robot figure pounce on Joe and fasten its metal fingers about his throat, crushing vertebrae and flesh alike.

Oblivious to her own danger, she rushed to rescue her already dead husband, but the metal fingers were inflexible. Belatedly she abandoned the attempt and ran into the hallway to the phone.

When the police arrived, they found her slumped against the wall in the hallway. She pointed toward the open doorway of the study, without speaking.

The police rushed into the study. At once there came the sounds of shots. Dozens of them, it seemed. Later both policemen admitted that they lost their heads and fired until their guns were empty.

But it was not yet the end of Adam.

It would perhaps be impossible to conceive the full horror of his last hours, but we can at least make a guess. Asleep when the boys entered the study, he awakened to a world he had never before perceived except very vaguely and under the soporific veil of opiate.

But it was a world vastly different even than that. There is no way of knowing what he saw—probably blurred ghostly figures, monstrous beyond the ability of his mind to grasp, for his eyes were adjusted only to the series of prisms and lenses that enabled him to see and coördinate the images brought to him through the eyes of the robot.

He saw these impossible figures, he felt pain and torture that were not of the flesh as he knew it, but of the spirit; agony beyond agony administered by what he could only believe were fiends from some nether hell.

And then, abruptly, as ten-year-old Paul shoved his head back into the helmet, the world he had come to believe was reality returned. It was as though he had returned to the body from some awful pit of hell, with the soul sickness still with him.

Before him he saw four human-like figures of reality, but beings unlike the only two he had ever seen. Smaller, seeming to be a part of the unbelievable nightmare he had been in. Two of them fled, two were within his grasp.

Perhaps he didn't know what he was doing when he killed Paul and Bill. It's doubtful if he had the ability to think at all then, only to tremble and struggle in his pitiful little rat body, with the automatic mechanisms of the robot acting from those frantic motions.

But it is known that there were three hours between the deaths of the two boys and the entry of Dr. MacNare at twelve thirty, and during those three hours he would have had a chance to recover, and to think, and to partially rationalize the nightmare he had experienced in realms outside what to him was the world of reality.

Adam must certainly have been calm enough, rational enough, to recognize Dr. MacNare when he entered the study at twelve thirty.

Then why did Adam deliberately kill Joe by breaking his neck? Was it because, in that three hours, he had put together the evidence of his senses and come to the realization that he was not a man but a rat?

It's not likely. It is much more likely that Adam came to some aberrated conclusion dictated by the superstitious feelings that had grown so strongly into his strange and unique existence, that dictated he must kill Joseph.

For it would have been impossible for him to have realized that he was only a rat. You see, Joseph MacNare had taken great care that Adam never, in all his life, should see *another* rat.

There remains only the end of Adam to relate.

Physically it can be only anticlimactic. With his metal body out of commission from a dozen or so shots, two of which destroyed the robot extensions of his eyes, he remained helpless until the coroner carefully removed him.

To the coroner he was just a white rat, and a strangely helpless one, unable to walk or stand as rats are supposed to. Also a strangely vicious one, with red little beads of eyes and lips drawn back from sharp teeth the same as some rabid wild animal.

The coroner had no way of knowing that somewhere in that small, menacing form there was a noble but lost mentality that knew itself as Adam, and held thoughts of a strange and wonderful realm of peace and splendor beyond the grasp of the normal physical senses.

The coroner could not know that the erratic motions of that small left front foot, if connected to the proper mechanisms, would have been audible as, perhaps, a prayer, a desperate plea to whatever lay in the Great Beyond to come down and rescue its humble creature.

"Vicious little bastard," the coroner said nervously to the homicide men gathered around Dr. MacNare's desk.

"Let me take care of it," said one of the detectives.

"No," the coroner answered. "I'll do it."

Quickly, so as not to be bitten, he picked Adam up by the tip of the tail and slammed him forcefully against the top of the desk.

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